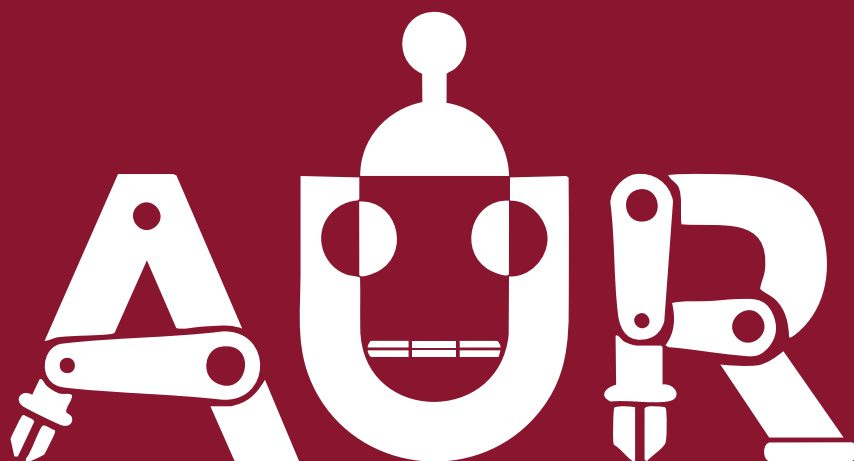




Training '26

Electrical | Phase I

Workshop 4: Communication Protocols
TRAINEE'S GUIDE



Introduction

This workshop focuses on configuring multi-controller architectures and implementing communication protocols between microcontrollers – specifically Arduino development boards – covering:

- Inter-board **UART Serial Communication** (Master-Slave Link).
- Synchronous **I2C Interface** for external displays.
- Data forwarding and real-time text rendering.
- Common ground references and hardware serial synchronization.

The objective is to gain practical experience in transferring data streams across two microcontrollers and rendering the received information on an I2C display.

Requirements

The workshop features a multi-node communication system subdivided into two core milestones. The main objective is to design and assemble a **Master-Slave Serial Bridge with I2C Display** circuit.

The following components will be provided to you in **Tinkercad Circuits**:

- 2× Microcontroller Boards (Arduino Uno)
- 1× I2C LCD Display (16x2 with PCF8574 Adapter)
- Breadboard & Virtual Jumper Wires

Please use **Tinkercad Circuits** or **Arduino IDE** to assemble the circuit and write the C++ firmware.

What follows are the set of required features and their descriptions.

1 Part 1: Arduino-to-Arduino UART Serial Bridge

Set up an asynchronous serial link between **Arduino 1 (Master)** and **Arduino 2 (Slave)**:

- Connect Master TX (Pin 1) → Slave RX (Pin 0).
- Connect Master RX (Pin 0) → Slave TX (Pin 1).
- Connect Master GND → Slave GND (Shared Common Ground).

Configure both hardware UART interfaces at a **Baud Rate of 9600 bps**.

Write firmware for **Arduino 1 (Master)** to read incoming text data from the computer's Serial Monitor and forward it immediately across the hardware serial line to **Arduino 2 (Slave)**.

❗ Important

Always ensure a common GND connection between both Arduino boards. Without a shared 0V reference, signal levels will be misread, causing data corruption.

💡 Tip

Avoid using `delay()` inside serial reading loops to prevent hardware buffer overruns.

2 Part 2: I2C LCD Display Rendering on Slave Node

Connect the 16x2 I2C LCD Display exclusively to **Arduino 2 (Slave)**:

- LCD SDA → Slave Analog Pin A4
- LCD SCL → Slave Analog Pin A5
- LCD VCC → Slave 5V
- LCD GND → Slave GND

Firmware requirements for **Arduino 2 (Slave)**:

1. Initialize the I2C LCD screen at address 0x27 (or 0x3F) using `<Wire.h>` and `<LiquidCrystal_I2C.h>`.
2. Listen continuously on the UART RX pin for incoming messages sent by **Arduino 1 (Master)**.
3. Parse the incoming string and render the received text live on the I2C LCD screen.

Tip

Use `lcd.clear()` or manage cursor positioning properly before printing new serial frames to prevent overlapping old characters on the display.

Appendix

1 Hardware Interfacing & Pinout Reference

1.1 UART Interface Connections

- **Data Rate:** 9600 bps
- **Logic Voltage:** 5V
- **Wiring Topology:** Cross-connected (TX to RX, RX to TX)

1.2 I2C Display Connections

- **Address:** 0x27 or 0x3F
- **Signal Lines:** SDA (Pin A4), SCL (Pin A5)

2 Arduino Uno Hardware Details

2.1 Power & Logic Properties

- Logic Level: 5V
- Output Voltages: 3.3V, 5V
- Common Ground: Shared GND pin connection required across both boards.

2.2 Pinout Diagram

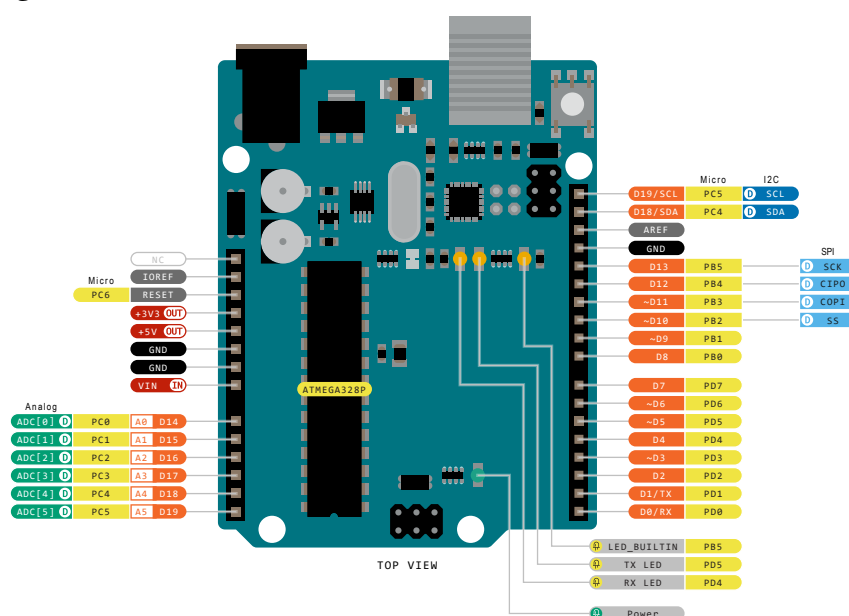


Figure 1: Arduino Uno REV3 Pinout Diagram